

```
In [14]: import pandas as pd      # importing libraries in python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [15]: data = pd.read_csv("C:/Users/ASUS/OneDrive/Desktop/EDA/dataset_olympics.csv")
data
```

Out[15]:

	ID	Name	Sex	Age	Height	Weight	Team	N
0	1	A Dijiang	M	24.0	180.0	80.0	China	C
1	2	A Lamusi	M	23.0	170.0	60.0	China	C
2	3	Gunnar Nielsen Aaby	M	24.0	NaN	NaN	Denmark	I
3	4	Edgar Lindenau Aabye	M	34.0	NaN	NaN	Denmark/ Sweden	I
4	5	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	F
...	...	...	...	...	...	...	...	...
69995	35656	Stuart Fitzsimmons	M	19.0	NaN	NaN	Great Britain	C
69996	35656	Stuart Fitzsimmons	M	19.0	NaN	NaN	Great Britain	C
69997	35657	David Thomas "Dave" Fitzsimons	M	26.0	170.0	65.0	Australia	J
69998	35657	David Thomas "Dave" Fitzsimons	M	30.0	170.0	65.0	Australia	J
69999	35658	Marek Fiurek	M	23.0	180.0	68.0	Czech Republic	C
70000	rows × 15	columns						

```
In [16]: data.head() # to get few information
```

```
Out[16]:
```

	ID	Name	Sex	Age	Height	Weight	Team	NOC	Games
0	1	A Dijiang	M	24.0	180.0	80.0	China	CHN	1992 Summer
1	2	A Lamusi	M	23.0	170.0	60.0	China	CHN	2012 Summer
2	3	Gunnar Nielsen Aaby	M	24.0	NaN	NaN	Denmark	DEN	1920 Summer
3	4	Edgar Lindenau Aabye	M	34.0	NaN	NaN	Denmark/ Sweden	DEN	1900 Summer
4	5	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	NED	1988 Winter

```
In [17]: data.info() # to get the basic details of data set
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 70000 entries, 0 to 69999
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   ID           70000 non-null  int64
1   Name         70000 non-null  object
2   Sex          70000 non-null  object
3   Age          67268 non-null  float64
4   Height       53746 non-null  float64
5   Weight       52899 non-null  float64
6   Team         70000 non-null  object
7   NOC          70000 non-null  object
8   Games        70000 non-null  object
9   Year         70000 non-null  int64
10  Season       70000 non-null  object
11  City         70000 non-null  object
12  Sport        70000 non-null  object
13  Event        70000 non-null  object
14  Medal        9690 non-null   object
dtypes: float64(3), int64(2), object(10)
memory usage: 8.0+ MB
```

```
In [18]: data.describe() # to attain statistical information
```

Out[18]:

	ID										Age																		
count	7	0	0	0	0	0	0	0	0	0	6	7	2	6	8	0	0	0	0	0	5	3	7	4	6	0	0	0	0
mean	1	8	0	8	1	8	4	6	9	8	6			2	5	6	4	4	6	4	5		1	7	5	5	0	5	3
std	1	0	2	3	5	6	1	3	2	5	3			6	4	8	5	2	3	9		1	0	3	8	4	2		
min					1	0	0	0	0	0	0			1	1	0	0	0	0	0	0		1	2	7	0	0	0	0
2 5 %					9	3	2	5	7	5	0	0	0	0	2	1	0	0	0	0	0		1	6	8	0	0	0	0
5 0 %	1	8	0	3	2	0	0	0	0	0	0			2	5	0	0	0	0	0	0		1	7	5	0	0	0	0
7 5 %	2	6	9	7	8	0	0	0	0	0	0			2	8	0	0	0	0	0	0		1	8	3	0	0	0	0
max	3	5	6	5	8	0	0	0	0	0	0			8	8	0	0	0	0	0	0		2	2	3	0	0	0	0

In [19]: `data.describe(include=["object"])`

Out[19]:

	Name					Sex	Team					NOC					Games					Season			
count	7	0	0	0	0	7	0	0	0	0	7	0	0	0	0	7	0	0	0	0	7	0	0	0	0
unique	3	5	5	5	6	2	8	2	7		2	2	6		5	1					2				
top	Oksana Aleksandrovnna Chusovitina					M	United States					USA					2 0 1 6 Summer					Summer			
freq	2 9					5 1	8 7 7	4 9 7 9					5 2 1 6					3 6 7 5					5 8 4 6 7		

In [20]: `data.isna().sum()` *# to get missing values*

Out[20]: ID 0
Name 0
Sex 0
Age 2732
Height 16254
Weight 17101
Team 0
NOC 0
Games 0
Year 0
Season 0
City 0
Sport 0
Event 0
Medal 60310
dtype: int64

In [21]: `data.duplicated().sum()` *# to get the duplicate values*

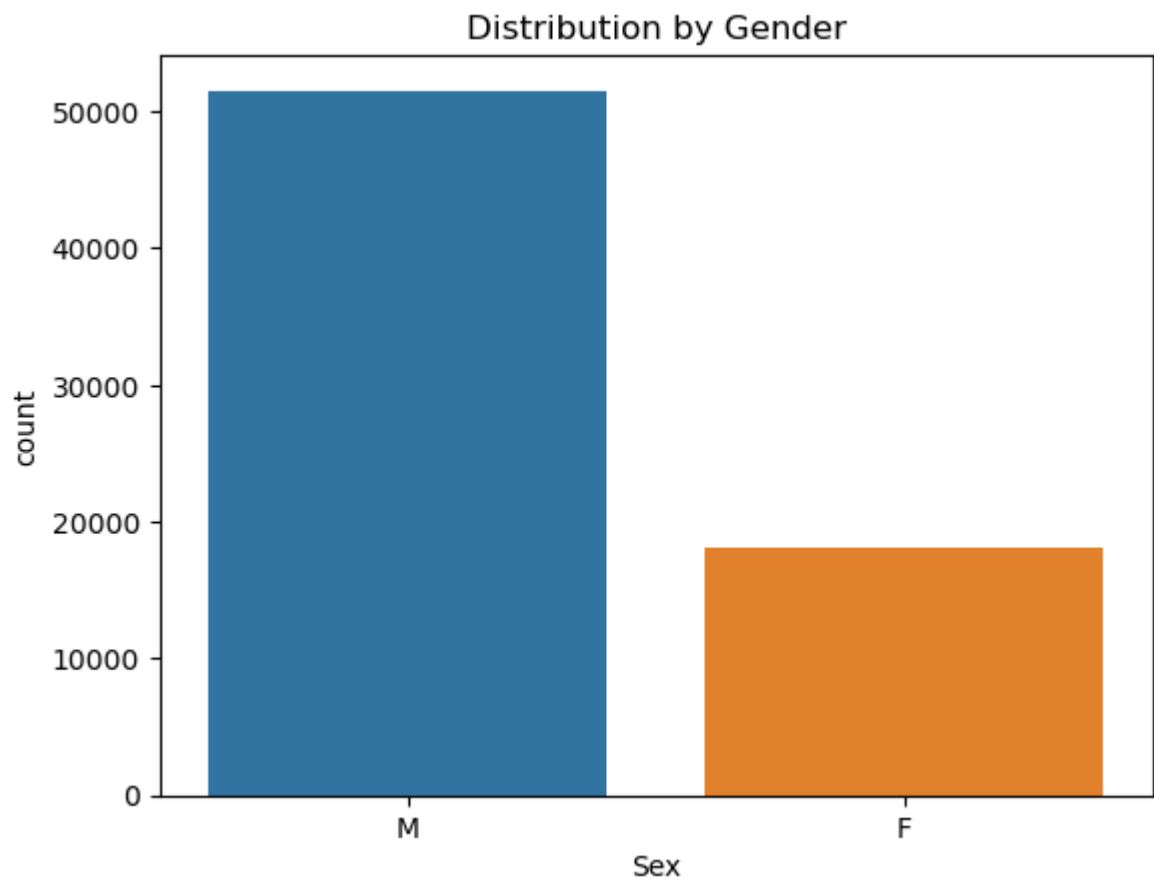
Out[21]: `np.int64(383)`

In [23]: `data.drop_duplicates(inplace=True)`

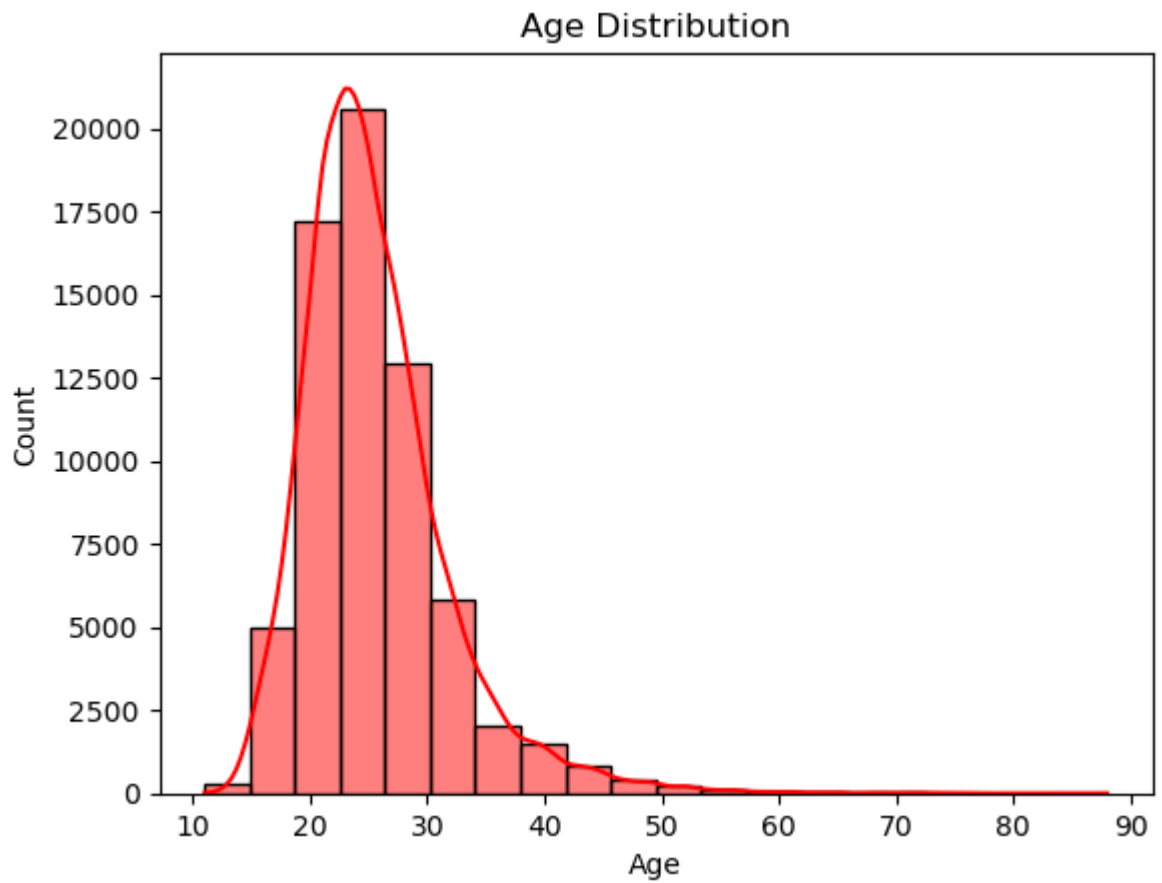
In [24]: `data.duplicated().sum()`

```
Out[24]: np.int64(0)
```

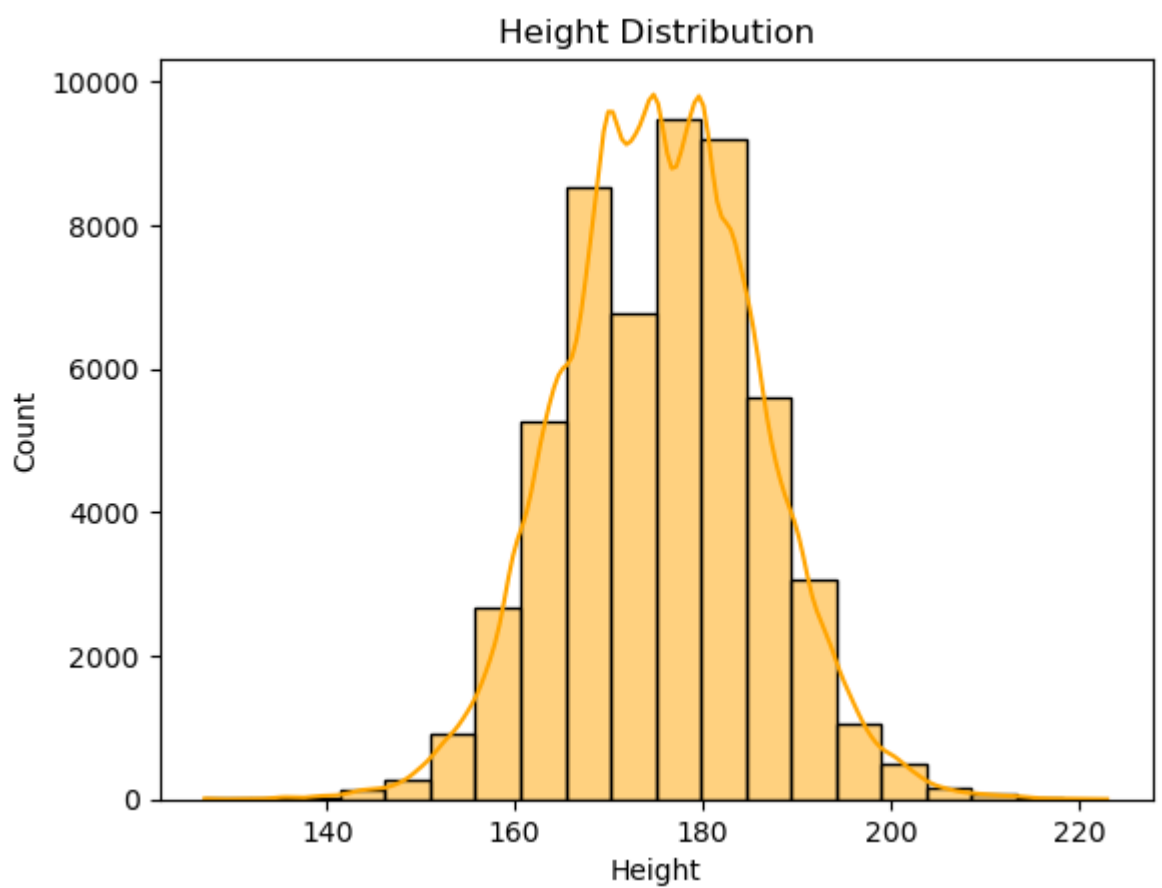
```
In [25]: # visualising
sns.countplot(data=data, x="Sex", hue="Sex")
plt.title("Distribution by Gender")
plt.show()
```



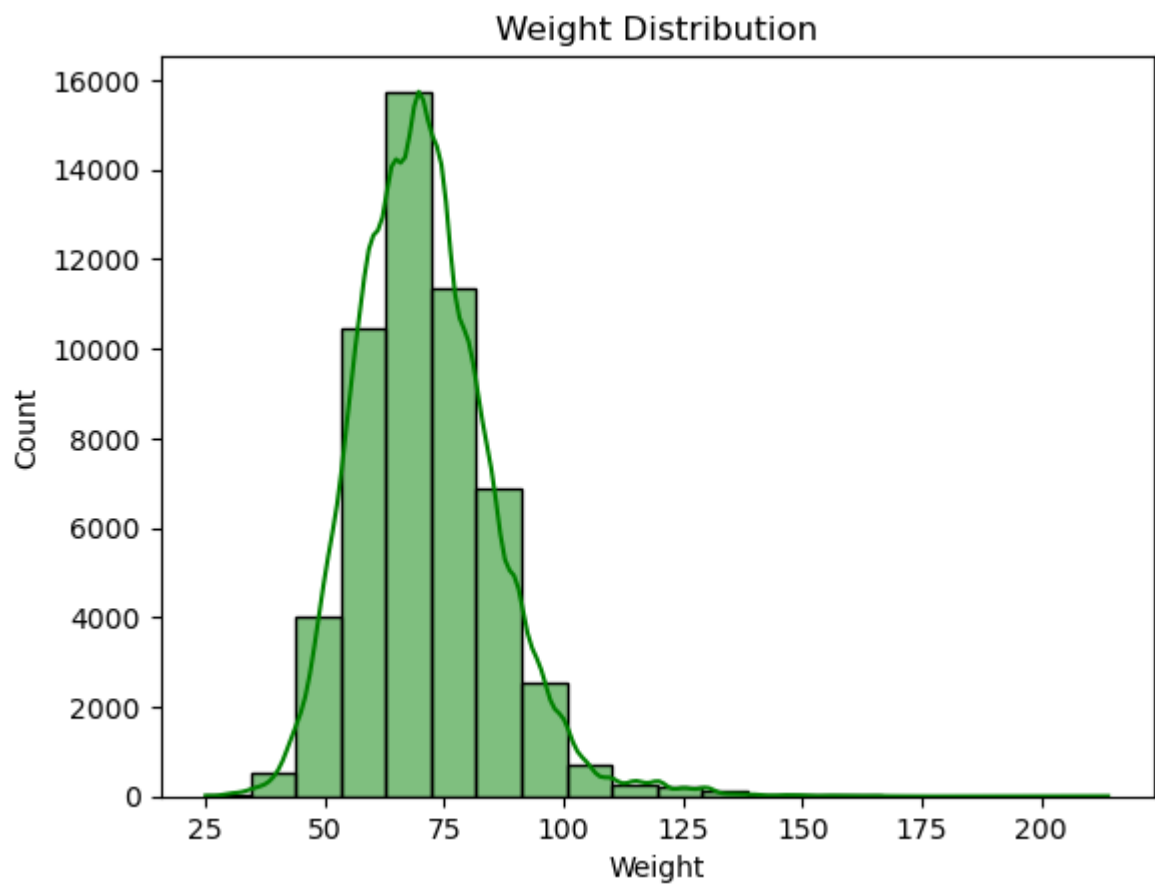
```
In [26]: sns.histplot(data=data, x="Age", color="red", bins=20, kde=True)
plt.title("Age Distribution")
plt.show()
```



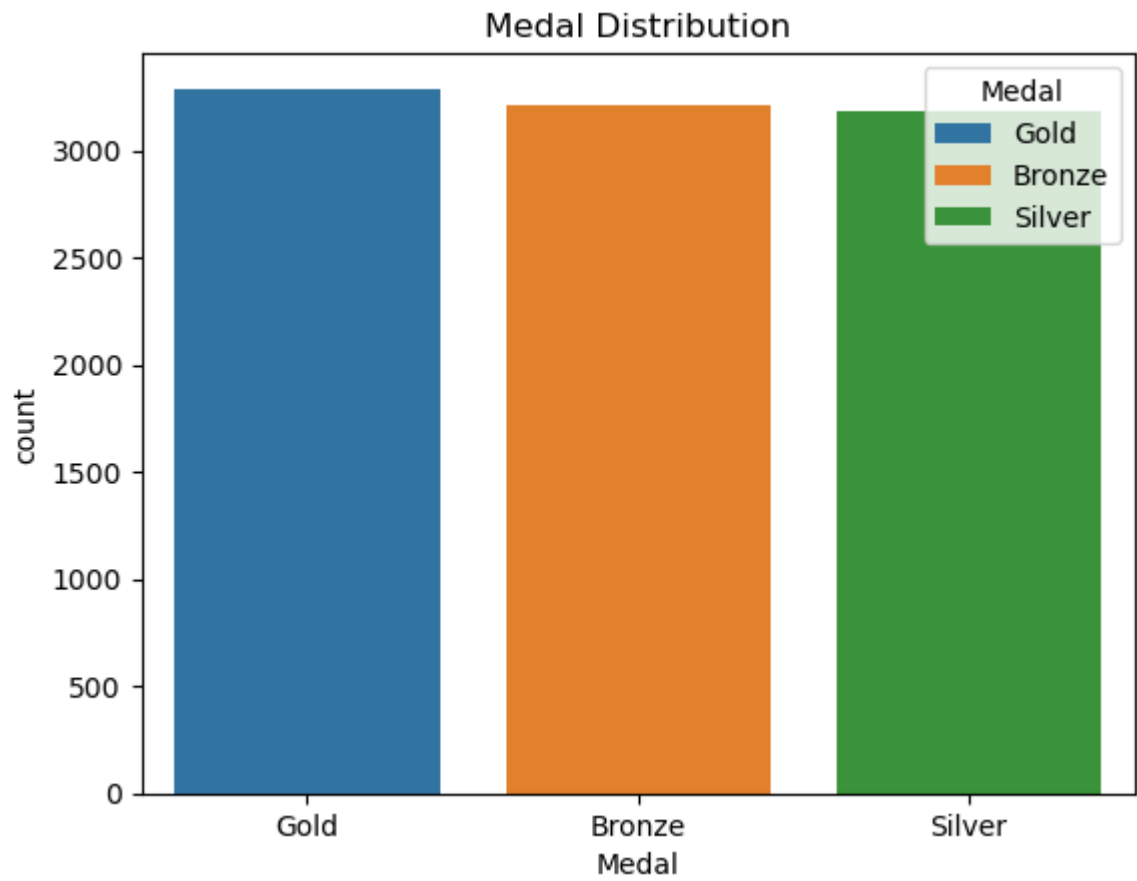
```
In [27]: sns.histplot(data=data,x="Height",color="Orange",bins=20,kde=True)  
plt.title("Height Distribution")  
plt.show()
```



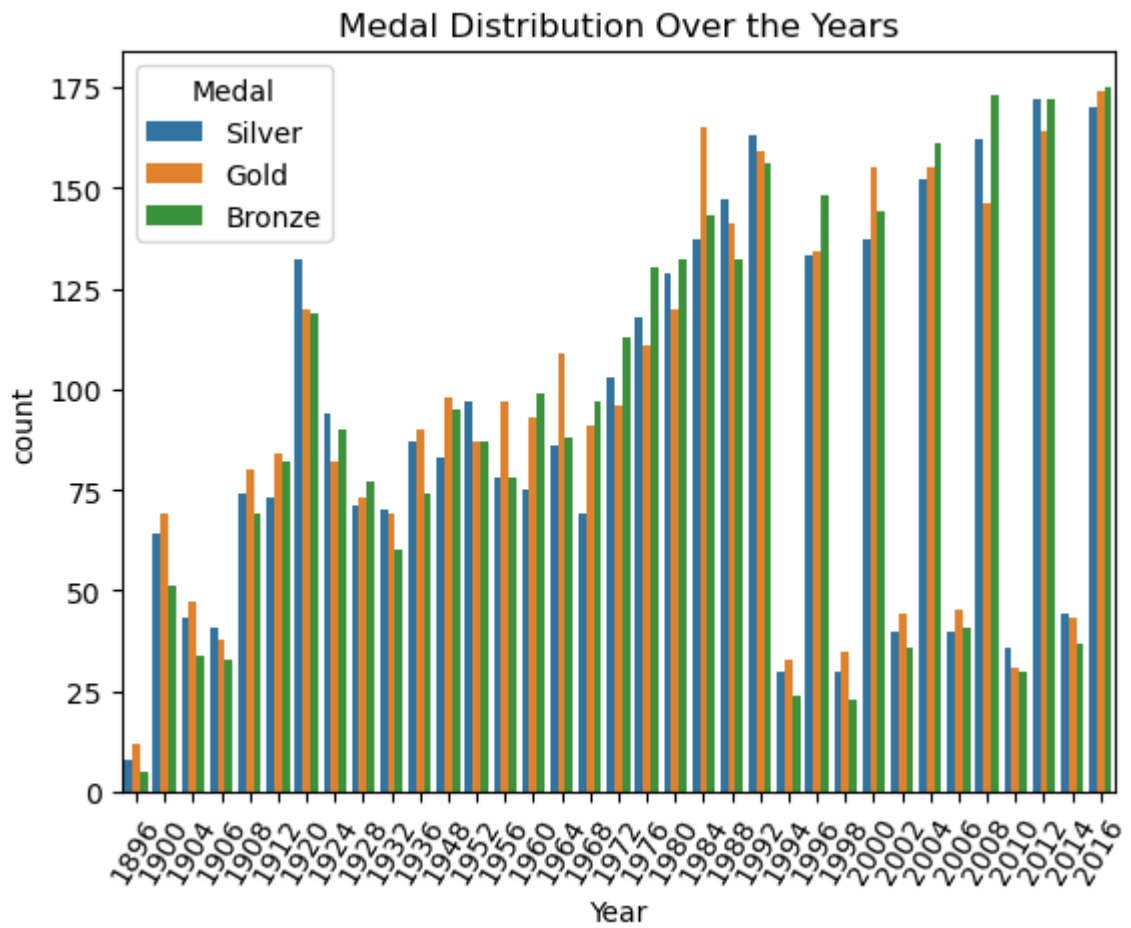
```
In [28]: sns.histplot(data=data,x="Weight",color="green",bins=20,kde=True)
plt.title("Weight Distribution")
plt.show()
```



```
In [29]: sns.countplot(data=data,x="Medal",hue="Medal")
plt.title("Medal Distribution")
plt.show()
```



```
In [30]: sns.countplot(data=data,x="Year",hue="Medal")
plt.title("Medal Distribution Over the Years")
plt.xticks(rotation=60)
plt.show()
```



```
In [31]: year_avg_age=data.groupby("Year")["Age"].mean()
print(year_avg_age)
```



```

Year
1896    23.029412
1900    29.119883
1904    27.063241
1906    26.989474
1908    27.000000
1912    27.965552
1920    29.241135
1924    28.252267
1928    27.973564
1932    29.606987
1936    27.245665
1948    28.363170
1952    26.273684
1956    26.316156
1960    25.136156
1964    24.852107
1968    24.316722
1972    24.126448
1976    23.656820
1980    23.312364
1984    24.060328
1988    24.257374
1992    24.637827
1994    24.487516
1996    25.338210
1998    25.143860
2000    25.435177
2002    26.029095
2004    25.780111
2006    26.091716
2008    25.685148
2010    26.150776
2012    25.993485
2014    26.082814
2016    26.259592
Name: Age, dtype: float64

```

```

In [32]: sport_median_height=data.groupby("Sport")["Height"].median()
         print(sport_median_height)

```

```

Sport
Alpine Skiing    173.0
Alpinism         NaN
Archery          172.0
Art Competitions 183.0
Athletics        176.0
...
Tug-Of-War       182.0
Volleyball       187.5
Water Polo       185.0
Weightlifting    168.0
Wrestling        172.0
Name: Height, Length: 65, dtype: float64

```

```

In [33]: sport_median_height=data.groupby("Sport")["Height"].median()
         print(sport_median_height.max())

```

```

190.0

```

```
In [34]: sport_median_height[sport_median_height==190.0]
```

```
Out[34]: Sport
Basketball    190.0
Name: Height, dtype: float64
```

```
In [35]: sport_median_height=data.groupby("Sport")["Height"].median()
print(sport_median_height.min())
```

```
164.0
```

```
In [36]: sport_median_height[sport_median_height==164.0]
```

```
Out[36]: Sport
Gymnastics    164.0
Name: Height, dtype: float64
```

```
In [37]: country_gender_count=data.groupby(["NOC", "Sex"])["ID"].count()
print(country_gender_count)
```

```
NOC  Sex
AFG  M      38
AHO  F       6
      M      27
ALB  F       4
      M       7
      ...
YUG  M     455
ZAM  F       3
      M      40
ZIM  F      41
      M      47
Name: ID, Length: 432, dtype: int64
```

```
In [38]: country_gold_medals=data[data["Medal"]=="Gold"].groupby("NOC")["Medal"].c
print(country_gold_medals)
```

```
NOC
ALG      1
ANZ      7
ARG     25
ARM      1
AUS     98
      ...
URU     13
USA    747
UZB      4
YUG     31
ZIM      7
Name: Medal, Length: 84, dtype: int64
```

```
In [39]: country_gold_medals=data[data["Medal"]=="Gold"].groupby("NOC")["Medal"].c
print(country_gold_medals.max())
```

```
747
```

```
In [40]: country_gold_medals[country_gold_medals==747]
```

```
Out[40]: NOC
USA     747
Name: Medal, dtype: int64
```

```
In [41]: country_gold_medals=data[data["Medal"]=="Gold"].groupby("NOC")["Medal"].count()
print(country_gold_medals.min())
```

1

```
In [42]: country_gold_medals[country_gold_medals==1]
```

```
Out[42]: NOC
ALG      1
ARM      1
AZE      1
CIV      1
COL      1
DOM      1
GEO      1
IOA      1
IRI      1
JOR      1
LUX      1
MGL      1
NEP      1
POR      1
THA      1
UAE      1
UGA      1
Name: Medal, dtype: int64
```

```
In [43]: sport_gender_avg_weight=data.groupby(["Sport","Sex"])["Weight"].mean()
print(sport_gender_avg_weight)
```

```
Sport      Sex      Weight
Alpine Skiing  F      62.154589
              M      77.725309
Alpinism      F           NaN
              M           NaN
Archery       F      61.023256
...
Water Polo    M      87.584973
Weightlifting F      66.189474
              M      79.927852
Wrestling     F      58.169014
              M      77.256240
Name: Weight, Length: 114, dtype: float64
```

```
In [44]: sport_gender_avg_weight=data.groupby(["Sport","Sex"])["Weight"].mean()
print(sport_gender_avg_weight["Wrestling"])
```

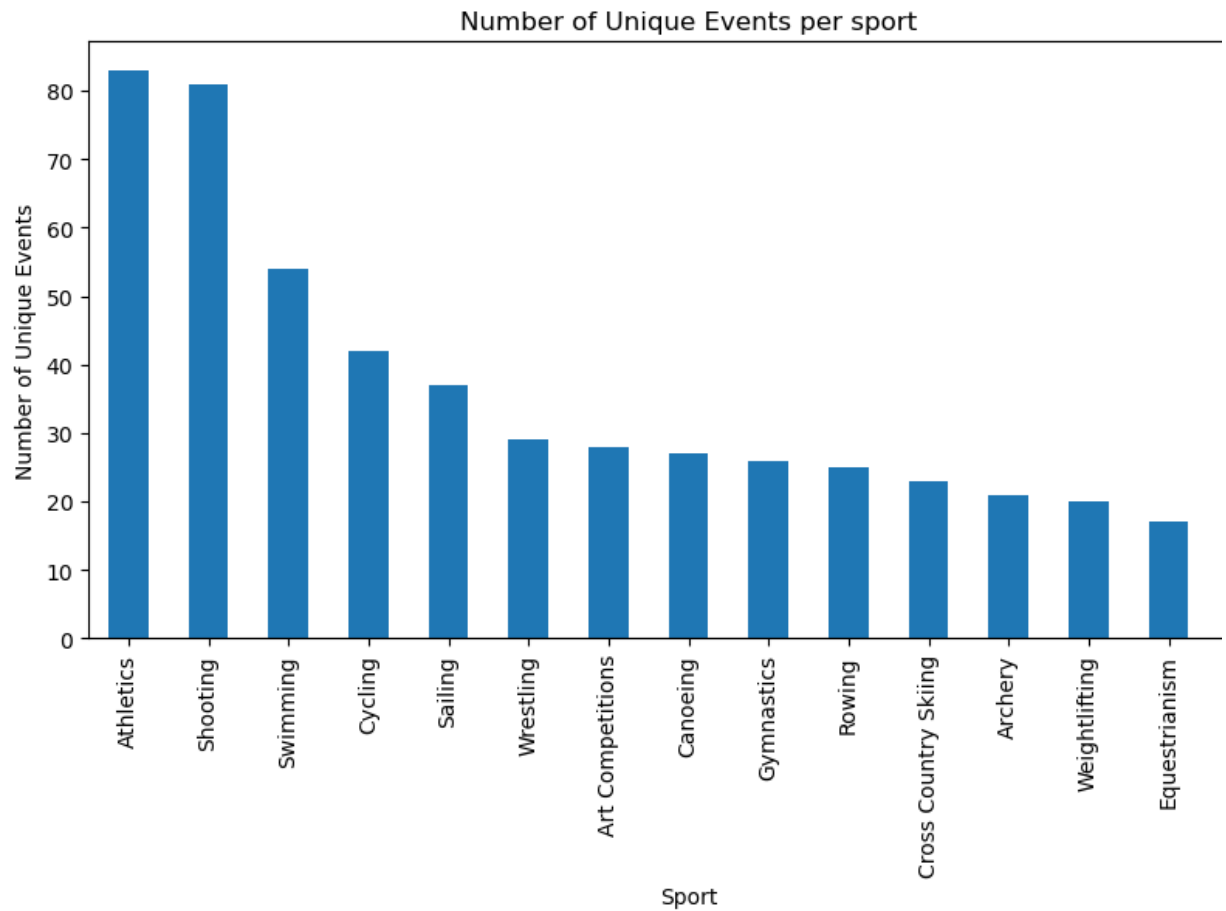
```
Sex
F      58.169014
M      77.256240
Name: Weight, dtype: float64
```

```
In [45]: sport_gender_avg_weight=data.groupby(["Sport","Sex"])["Weight"].mean()
print(sport_gender_avg_weight["Wrestling"]["F"])
```

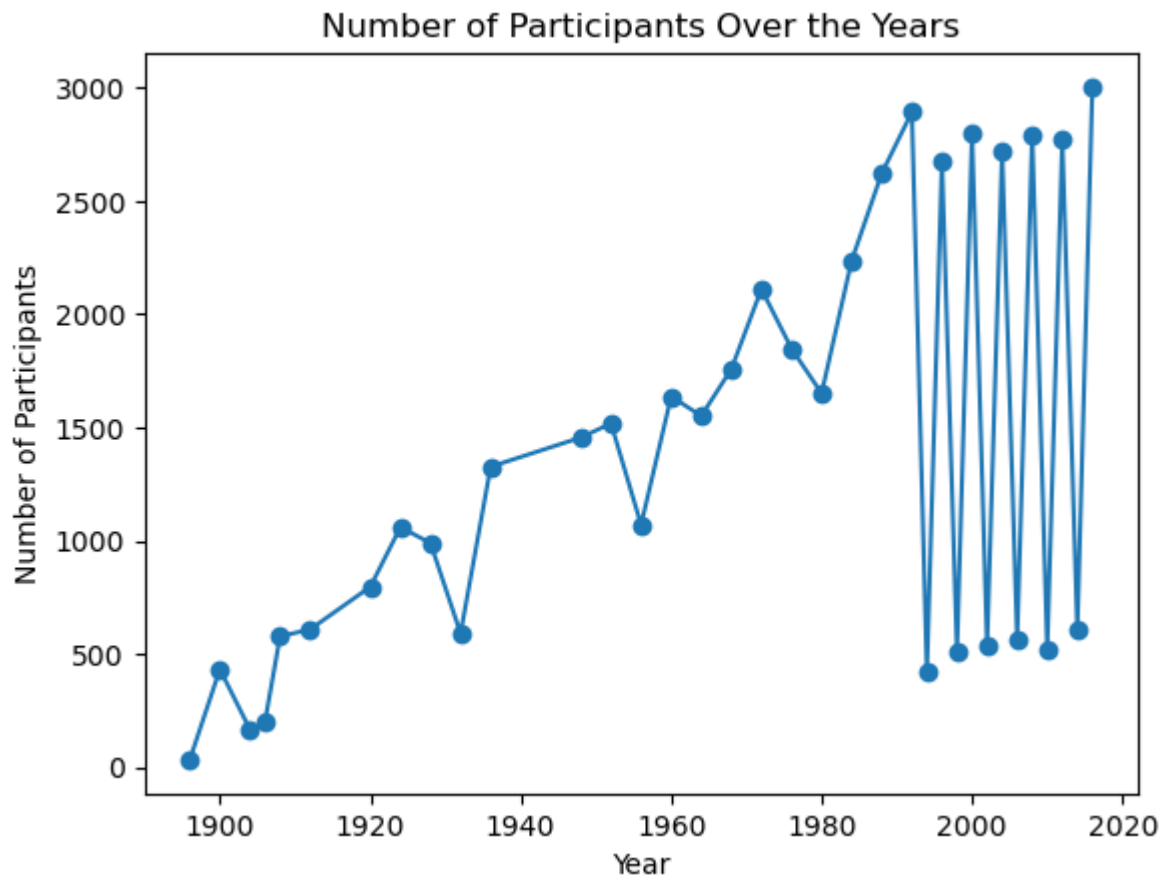
58.16901408450704

```
In [46]: sport_event_count=data.groupby("Sport")["Event"].nunique().sort_values(ascending=True)
sport_event_count.head(15).plot(kind="bar",figsize=(10,5),rot=45)
plt.title("Number of Unique Events per sport")
plt.xlabel("Sport")
```

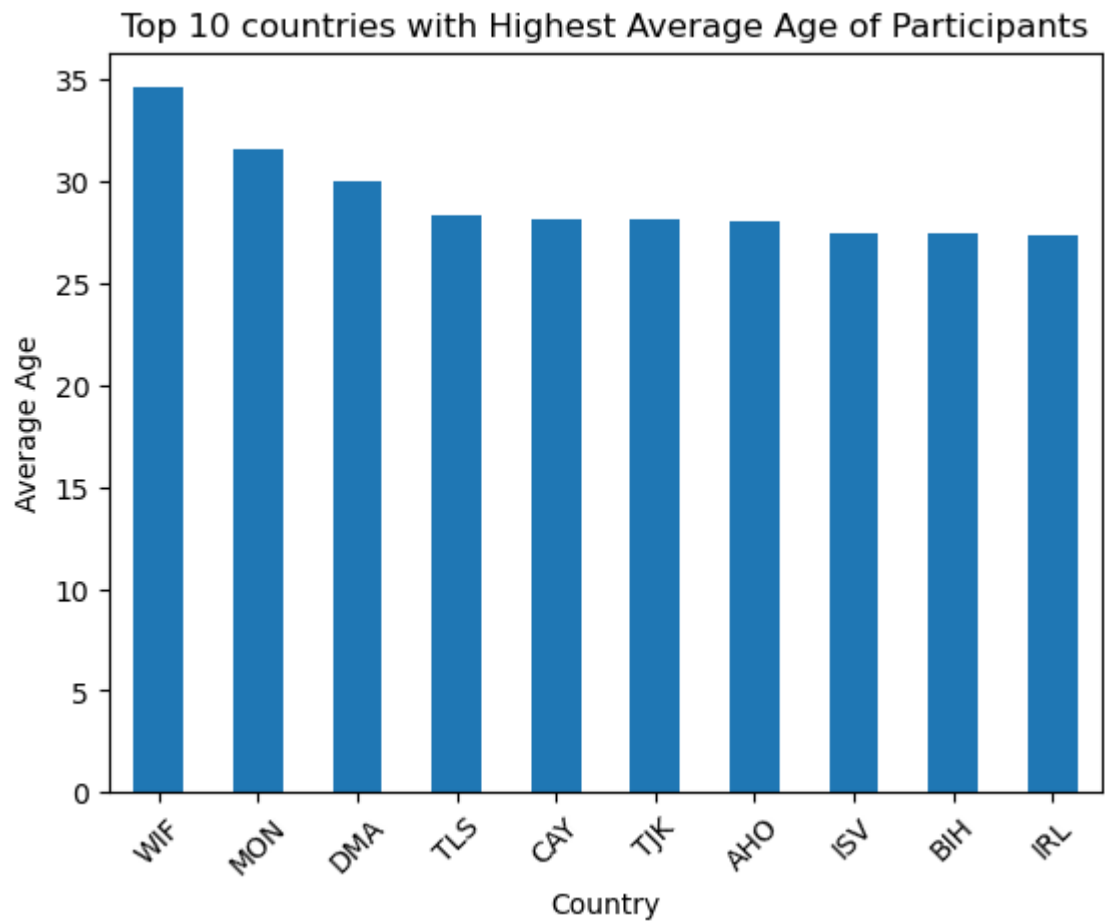
```
plt.ylabel("Number of Unique Events")
plt.xticks(rotation=90)
plt.show()
```



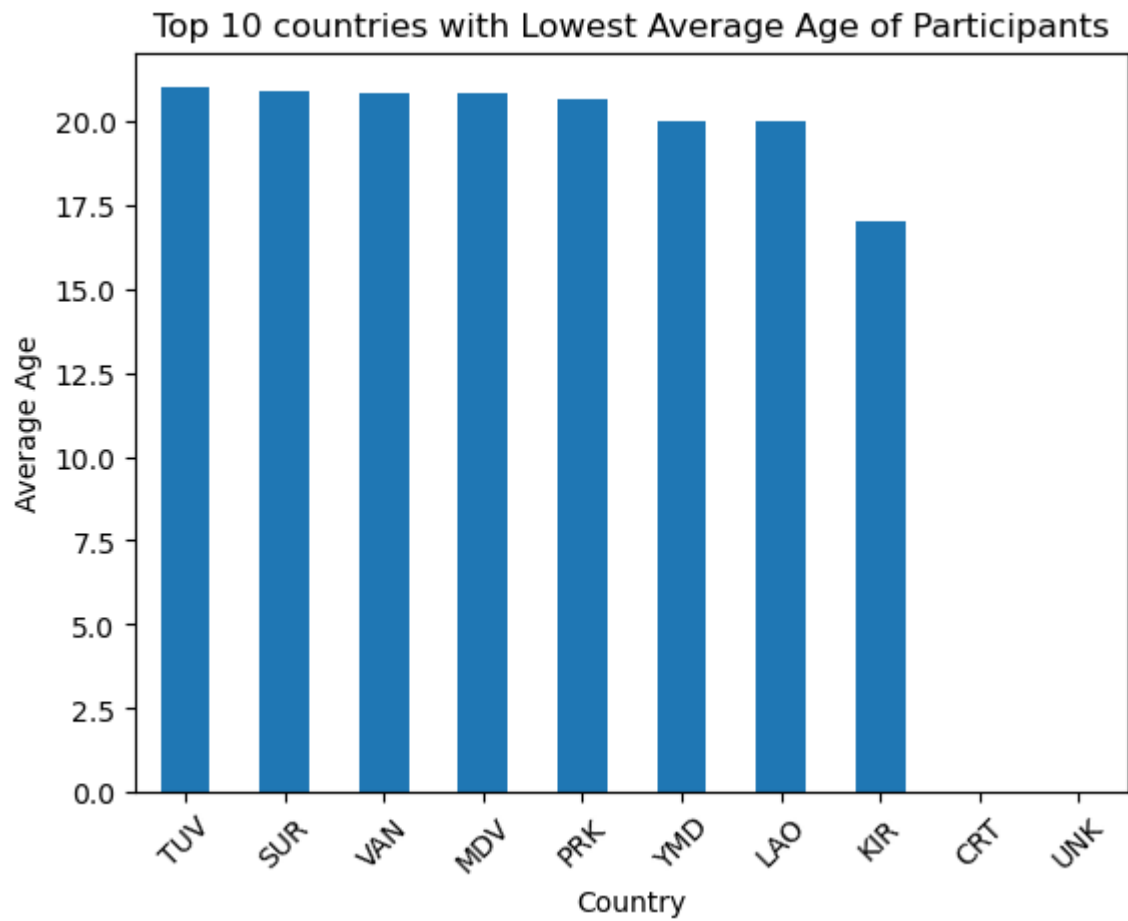
```
In [47]: year_participant_count=data.groupby("Year")["ID"].nunique()
year_participant_count.plot(kind="line",marker="o")
plt.title("Number of Participants Over the Years")
plt.xlabel("Year")
plt.ylabel("Number of Participants")
plt.show()
```



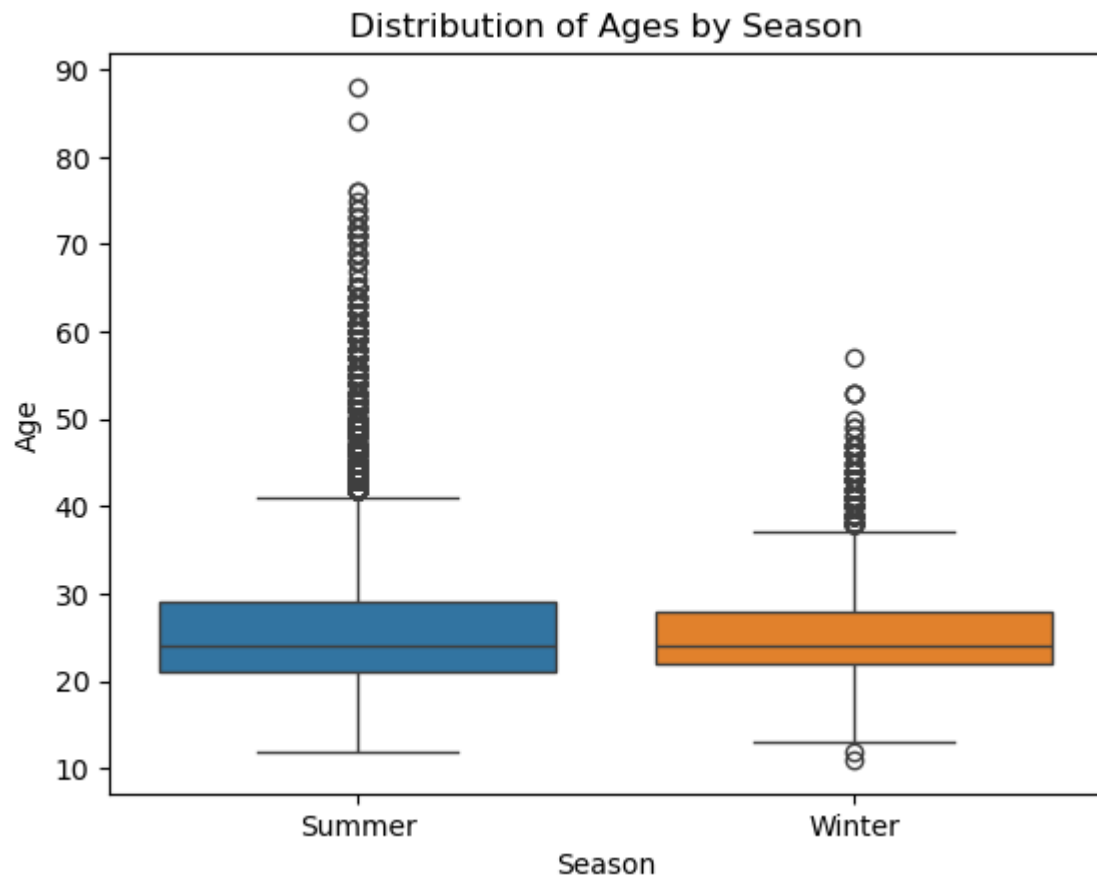
```
In [48]: country_avg_age=data.groupby("NOC")["Age"].mean().sort_values(ascending=F
country_avg_age.head(10).plot(kind="bar")
plt.title("Top 10 countries with Highest Average Age of Participants")
plt.xlabel("Country")
plt.ylabel("Average Age")
plt.xticks(rotation=45)
plt.show()
```



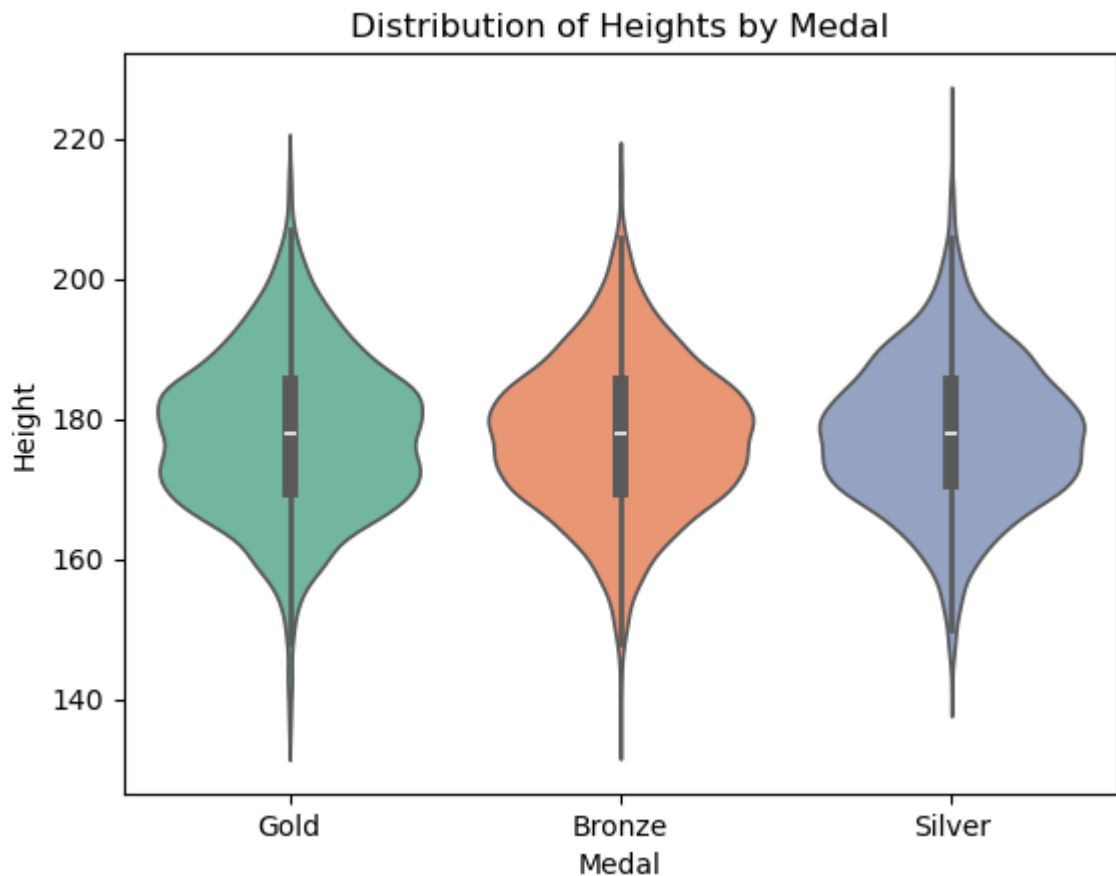
```
In [49]: country_avg_age=data.groupby("NOC")["Age"].mean().sort_values(ascending=False)
country_avg_age.tail(10).plot(kind="bar")
plt.title("Top 10 countries with Lowest Average Age of Participants")
plt.xlabel("Country")
plt.ylabel("Average Age")
plt.xticks(rotation=45)
plt.show()
```



```
In [50]: sns.boxplot(data=data, x="Season", y="Age", hue="Season")  
plt.title("Distribution of Ages by Season")  
plt.xlabel("Season")  
plt.ylabel("Age")  
plt.show()
```



```
In [51]: sns.violinplot(data=data,x="Medal",y="Height",hue="Medal",palette="Set2",  
plt.title("Distribution of Heights by Medal")  
plt.xlabel("Medal")  
plt.ylabel("Height")  
plt.show()
```

```
In [52]: most_medals_country=data["NOC"].value_counts().idxmax()
print("Most Medal_Winning Country: ", most_medals_country)
```

Most Medal_Winning Country: USA

```
In [53]: tallest_athlete=data[data["Height"]==data["Height"].max()]
print("Tallest Athlete:")
print(tallest_athlete[["ID","Name","Height","Sport"]])
```

Tallest Athlete:

ID	Name	Height	Sport
32376 16639	Tommy Loren Burleson	223.0	Basketball

```
In [54]: heaviest_athlete=data[data["Weight"]==data["Weight"].max()]
print("Heaviest athlete: ")
print(heaviest_athlete[["ID","Name","Weight","Sport"]])
```

Heaviest athlete:

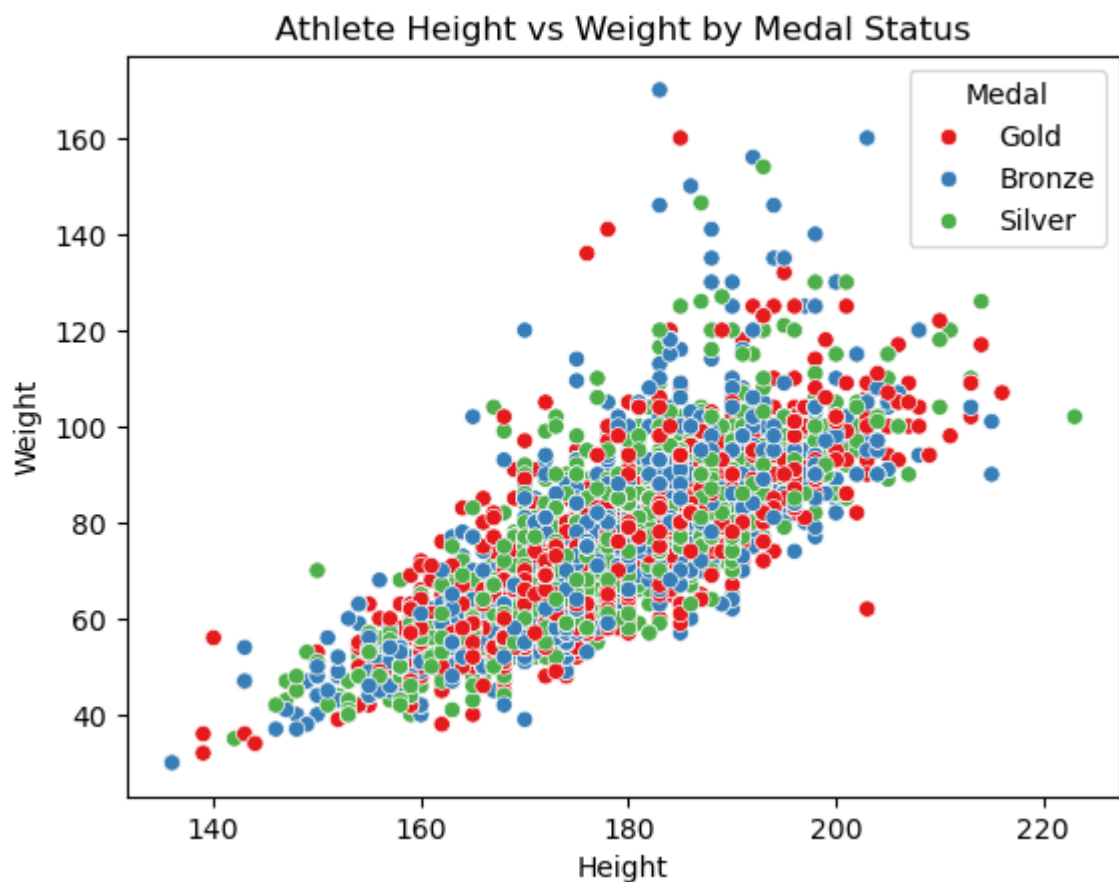
ID	Name	Weight	Sport
23155 12177	Ricardo Blas, Jr.	214.0	Judo
23156 12177	Ricardo Blas, Jr.	214.0	Judo

```
In [55]: heaviest_athlete=data[data["Weight"]==data["Weight"].min()]
print("Heaviest athlete: ")
print(heaviest_athlete[["ID","Name","Weight","Sport"]])
```

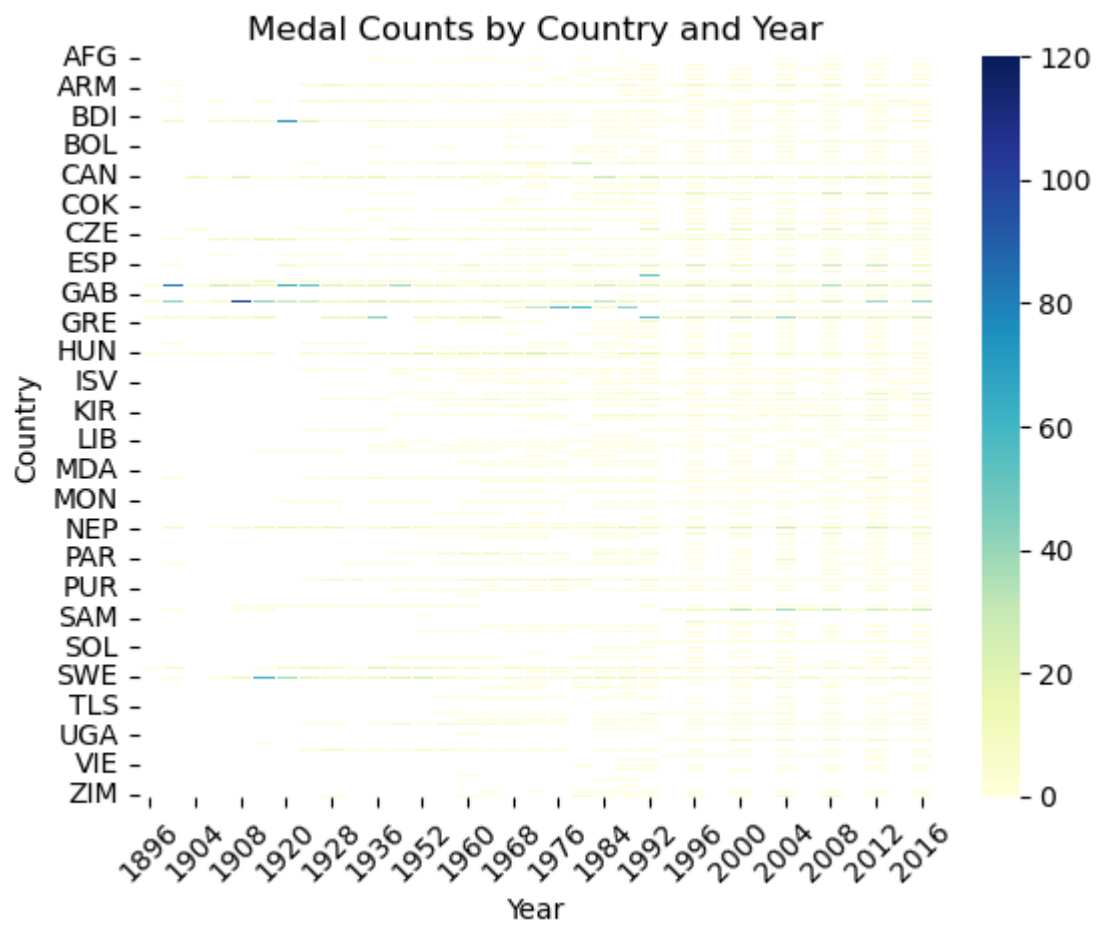
Heaviest athlete:

	ID	Name	Weight	Sport
40849	21049	Choi Myong-Hui	25.0	Gymnastics
40850	21049	Choi Myong-Hui	25.0	Gymnastics
40851	21049	Choi Myong-Hui	25.0	Gymnastics
40852	21049	Choi Myong-Hui	25.0	Gymnastics
40853	21049	Choi Myong-Hui	25.0	Gymnastics
40854	21049	Choi Myong-Hui	25.0	Gymnastics

```
In [56]: sns.scatterplot(data=data,x="Height",y="Weight",hue="Medal",palette="Set1")
plt.title("Athlete Height vs Weight by Medal Status")
plt.xlabel("Height")
plt.ylabel("Weight")
plt.legend(title="Medal")
plt.show()
```



```
In [57]: medals_by_country_year=data.pivot_table(index="NOC",columns="Year",values="Medal")
sns.heatmap(medals_by_country_year,cmap="YlGnBu",linewidths=0.5)
plt.title("Medal Counts by Country and Year")
plt.xlabel("Year")
plt.ylabel("Country")
plt.xticks(rotation=45)
plt.show()
```



In []: